



SJAA EPHEMERIS

SJAA Activities Calendar

Jim Van Nuland

October

- 7 **General meeting at Houge Park.** Dr. Steve Stahler of UC Berkeley will speak on "Why Are There Stars?". 8 p.m.
- 12 ATM Workshop at Houge Park. 7:30 p.m.
- 13 Astronomy Class at Houge Park. 7:30 p.m. Guest Speaker: Jim Albers about observing satellites and rocket launches.
- 13 Houge Park star party. Sunset 6:33 p.m., 47% moon rise 11:58 a.m. Star party hours: 7:30 to 10:30
- 14 Dark sky weekend. Sunset 6:32 p.m., 37% moon rise 1:02 a.m.
- 21 Dark sky weekend. Sunset 6:23 p.m., 0% moon rise 7:55 a.m.
- 27 Houge Park star party. Sunset 6:15 p.m., 32% moon sets 10:10 p.m. Star party hours: 7:00 to 10:00
- 28 ATM Workshop at Houge Park. 7:30 p.m.
- 29 DST ends. Retard clock. 2 am -> 1 am
- 31 Halloween. Set up at home for the little goblins

November

- 4 **General meeting at Houge Park.** 8 p.m.
- 5 **Fall Swap at Houge Park.** Noon.
- 9 ATM Workshop at Houge Park. 7:30 p.m.
- 10 Houge Park star party. Sunset 5:01 p.m., 64% moon rise 9:49 p.m. Star party hours: 7:00 to 10:00
- 10 Astronomy Class at Houge Park. 7:30 p.m.
- 11 Dark sky weekend. Sunset 5:00 p.m., 54% moon rise 10:53 p.m.
- 18 Dark sky weekend. Sunset 4:56 p.m., 1% moon rise 5:47 a.m.
- 24 Houge Park star party. Sunset 4:52 p.m., 19% moon sets 8:10 p.m. Star party hours: 7:00 to 10:00
- 25 ATM Workshop at Houge Park. 7:30 p.m.

The Board of Directors meets at 6:00 p.m. preceding each general meeting. All are welcome.

24 hour news and information hotline:
(408) 559-1221

October General Meeting

Steven Stahler: Why are there Stars?

On a clear night, the sky is filled with countless stars. These objects must be the natural outcome of processes that occur all the time, throughout our Galaxy and others. Indeed, images taken with radio and infrared telescopes show us new stars forming relatively nearby. The progenitor objects are large clouds permeating interstellar space. These clouds undergo gravitational collapse to form primitive stars, which then evolve to become mature objects like our own Sun. A large body of research, mostly undertaken within the past few decades, has led to a good understanding of the basic evolutionary process. Nevertheless, deep mysteries remain in this active and exciting field.

Our speaker Dr. Steven Stahler is an astrophysicist at U. C. Berkeley. Raised in Maryland, he attended graduate school at Berkeley in physics. He was a professor at MIT before returning to the Bay Area in 1992. His research centers on the problem of star formation, and he recently coauthored the first comprehensive textbook in the field ("The Formation of Stars," Stahler & Palla, Wiley-VCH, 2004). Trained as a theoretical physicist, Steve especially delights in the esthetic aspect of his research, which he tries to convey in his numerous public talks.

October 7 @ 8 p.m.
Houge Park



The SJAA group at the annual Yosemite trip. Photo dates July 21, 2006. Courtesy of Morris Jones.

Star-B-Q '06



Clockwise from above: Mike Koop, SJAA President, talks to Dr. Scott Sanford and Ken Frank to get help on the trivia contest (which Mike won); Dave North and Akkana Peck are quite impressed by Dr. Sanford's kids; Denni Medlock discusses the early days of the Fremont Peak Observatory; the chow line forms quickly which explains how all of these people got in front of the person with the camera (Paul Kohlmler).



Is It or Isn't It?

Akkana Peck

I'm sure you've heard by now about last month's IAU vote officially demoting Pluto. According to the IAU, Pluto is no longer a planet; it's a "dwarf planet".

The decision established three categories of bodies orbiting the sun:

1. Planets: objects which are large enough to clear their orbits of any other bodies (aside from moons, presumably). This includes the first eight planets from Mercury to Neptune.
2. Dwarf planets: objects which are large enough that gravitation makes them round, but not large enough to have cleared their orbits. This category includes Pluto, and also some of the larger non-planetary bodies such as Ceres and the recently discovered 2003 UB313.
3. Small Solar System Bodies: anything too small to be a dwarf planet.

The definition has some confusing holes. For example, many of the news articles describing the decision say that Pluto fails the "planet" test because its orbit overlaps with that of Neptune: therefore Pluto hasn't cleared its orbit and isn't a planet. But wouldn't that eliminate Neptune from the planet club as well?

I'm guessing that the news articles got that part wrong, and the non-orbit-clearing that disqualifies Pluto is actually the nearness of other Kuiper belt objects. That makes more sense, but it has holes too. Earth shares its orbit with quite a number of Earth-crossing asteroids – at least 300 have been identified, of course there are a lot more than that: one estimate is that there are around 1500 larger than a kilometer, and 135,000 larger than 100 meters. So by the new definition, Earth can't be a planet either.

Nor can Mars, since there are plenty of asteroids that veer over into Mars' orbit. And what about the thousands of Trojan asteroids sharing Jupiter's orbit? If

the new definition doesn't even include Jupiter as a planet, what good is it?

Not all scientists agree with the IAU's decision. Dr Alan Stern, leader of the New Horizons mission currently winging its way to Pluto, has been outspoken in his criticism of the decision – not just on scientific grounds, but on political ones (the vote was apparently held on the last day of the meeting, when many attendees were already on their way home; 424 attendees voted, which, Stern notes, represents less than 5% of the world's astronomers). More than 300 astronomers have signed a petition disagreeing with the IAU's definition, and vowing not to use it. Some textbook authors and museums are scurrying to make updates, but others are holding off, expecting a backlash.

I don't think we've heard the end of this yet. We may not get Pluto back as a planet; but I hope we at least can look forward to a clearer definition of exactly what constitutes a planet, one on which astronomers can agree. In the meantime, SJAA member Mark Taylor points out one up side: it just got a lot easier for an observer, regardless of location or telescope size, to see all the planets in a single night!

The object of all this controversy is sitting placidly at the foot of Ophiuchus, within reach of medium to large amateur scopes. It's already past the meridian by the time the sky gets dark, and is fairly far south so it never gets very high. There's still time to catch it if you start early, but it'll be a bit more work than it was a few months ago.

Wait – how can it be in Ophiuchus? That's not a sign of the Zodiac – don't planets move along the ecliptic and therefore stay within the Zodiacal constellations?

This is an example of the differences

between astronomy and astrology. The official boundaries of the astronomical constellations don't always match the boundaries of the zodiac, which is set up so that each zodiacal sign spans a roughly equal distance along the ecliptic, one sign per month. The astronomical constellation boundaries follow no such restriction. In fact, the constellation Ophiuchus occupies quite a bit more space along the ecliptic than its Zodiacal neighbor Scorpius does, and planets do pass through Ophiuchus' boundaries. (In fact, last month Pluto was just barely over the border from Ophiuchus into Serpens Cauda, which does not itself reach the ecliptic, but is close enough to it that Pluto's inclined orbit takes it inside the constellation boundaries.)

That said, an alert reader pointed out that I erred in past months in placing Uranus in Ophiuchus. Oops! It's actually in Aquarius. This month it's just south of lambda Aquarii, which should make it a snap to find in binoculars or a telescope. Neptune is in Capricornus, a bubble rising from the top of the "champagne glass". They're both well up by the time the sky gets dark, and transit a few hours before midnight, so they're ideally placed for early evening observers.

Saturn rises about an hour after midnight and will be available to morning observers.

Jupiter is still visible, barely, in the evening twilight, but as the month progresses it will disappear in the sun's glow. In the latter half of the month Mercury joins it.

Venus and Mars are lost in the sun's glare and are both in conjunction this month, Mars on the 23rd and Venus (superior conjunction) on the 27th. We won't see either of them again until December, when Venus will emerge low in the evening sky and Mars in the morning.

A Club Observatory, tale from San Diego

Steven Nelson

The summer of 2003 I took an evening “off” from my vacation to do some “astro business”, like visiting an observatory and talking astronomy! The Board had been discussing 50th year objectives for the SJAA. I was researching the issue by corresponding with other clubs that seem to have successful observing site operations. Here is what I found at the San Diego Astronomical Association’s site through the kind efforts of SDAA Observatory Director Jim Traweek <http://www.sdaa.org/index.cfm?fuseaction=tds.info>

The ten acre Tierra Del Sol site was purchased about 30 years ago when the SDAA had about 50 members. (Ed. Note: A site plan can be seen at http://ephemeris.sjaa.net/0610/SDAA_SitePlan.pdf). It is about 70 miles East of San Diego and the route can mainly be driven on interstate 15. This site is 10 level acres in an area of short chaparral bush (great horizons). The important site selection points seem to have been, 1 hour driving access, electric lines already in area, dirt road access is limited to the last 1/8 mile, and dark sky. Additionally, the area is near Indian lands with limited development potential. The large site has allowed slow but substantial development over several decades. The ‘one hour access rule’ means it is used monthly for club “public” star parties that are very well attended. Public visiting is encouraged at parties, use requires membership.

The site offers four major services to the SDAA, (1) a developed dark sky site for large star parties (user pads), (2) a 21” equatorially mounted scope (club observatory), (3) an area for about 68 individual small pads and parking (private pads), and (4) a place for six large member observatories (private observatories). The user pads are together in an open field near the restrooms, observatory, and a clubroom. These are available to members on a first-come basis. Two long cement strips for this use have space for back-in

parking and all have electric outlets. As the decades have passed there have been increases in members, and the usage and extent of these pads have increased. The observatory is always open and hosted at club functions and an attached club room provides a microwave, hot beverages, and work tables. There is an outside barbeque area. These areas are wheelchair accessed by gently-sloped cement ramps. These spaces provide plenty of room for the large group activities of the club.

The electric service seems important to their site. All telescope sites here have electricity. A well was drilled so the site has a regular water supply. This water, and an electric pump, allowed a restroom building with sinks, flush toilets, and even a modest shower with hot water! I think such amenities (about like a CA state campground) have helped their success over the decades. Large gatherings can use the site without totally roughing it. Females and families are more likely to frequent and use a place with restrooms!

Such a site has been the accumulation of efforts over the decades and generations of club members. Someone designed and installed a leach-field (for sewage disposal) 20 years ago, 5 years ago someone finished off the ‘club room’, and volunteers used rented equipment to clear bush and grade more land. Many people chip in volunteer effort and extra volunteer funds when major capital projects are needed.

MORE DETAILS

The sixty individual pads are small sites with electric service that are available for yearly lease. Serious observing can be done here in relative isolation. Places for about 10-15% of the club are available and the lease about doubles club yearly membership (to about \$70). This seems to be one of

the major ‘user fee’ sources of income that helps pay annual electricity, taxes and maintenance. Pad “development”, if desired, is the user responsibility. Typically users put in a cement pad and a permanent pier. Occasionally benches and equipment storage cabinets are installed. Some of the initial private pads have been with their ‘owners’ 30 years. The “lease” arrangement allows interest to be retained by the club without ‘license sale’ profiteering. There is a drawing of the observatory grounds at the SDAA web site.

The club observatory seemed undersubscribed. There is no extra charge, but training and a stint as a host docent for one-public-star-party-a-year are the only requirements. This is a facility about the size of the Fremont Peak State Park telescope.

The private observatories area is for those who could afford building a major observatory and help with the original land purchase. These sites are now built-out and the SDAA has planning for new sites. I believe these observatories sell privately, but require SDAA membership and some SDAA overview. The private observatories apparently pay for half the taxes (as a form of ground rent). The concept of ‘ground rent’ is fairly foreign to most Californians, but the situation of houses and businesses on land of Stanford University and Mission College are two local examples.

I think the large site in an easy-to-reach location helped to make this club venture a success. I was very surprised by the large infrastructure and the multiple uses. But space allows both expansion and ‘errors’ in plans. An expansion in ‘private pads’ allowed more members a space (and increased annual fees for the club). Still, about 30% of the site is unused after 30 years. This extra space seems to allow all interest groups and individuals room

Continued on page 5

to grow and to pursue their specialized interests.

There are several ways the costs are contained. The fund for projects seems to grow by donations when there is a lot of interest. Projects taking \$5000 and several hundred volunteer hours seem to take place only every few years. It really adds up over the decades! There are now over 500 members in the SDAA and the general dues are \$35. Some of this cash is used to support the group area and club observatory. As I mentioned, additional yearly fees are also collected for the private pads and observatories. The observatory web page is at <http://www.sdaa.org/index.cfm?fuseaction=tds.info>.

Coming articles will benchmark the Orange County Astronomers Anza and the Riverside Astronomical Association sites. The SJAA board is now in an active site campaign. The committee has focused on the dark-sky area south of Hollister, and 10-20 usable acres. They still are interested in rural land that might be suitable and available.

Editorial

In August, the IAU decided on a definition of a planet. Please read Akkana's article on page 4. The new definition removed Pluto from the list of planets. Many of my favorite astronomers, people like Alexei Filippenko and Neil deGrasse Tyson, are probably happy with that result. I have to agree that the new definition was better than the committee-created resolution which would have resulted in a planetary delta of +3 instead of -1. And I'm also glad that we now have a "scientific" definition of the word planet. But I want to argue for one small modification: the Pluto Exception.

I probably don't have to explain what I mean - make an exception for Pluto. Let the astronomers have their new definition of a planet but keep Pluto in

Silicon Valley Astronomy Lecture Series

Alex Filippenko on Dark Energy on October 4, 2006

7 p.m. at Foothill College

Andrew Fraknoi

Astronomer Alex Filippenko of the University of California, Berkeley will give a non-technical, illustrated talk on: "Dark Energy and the Runaway Universe" as part of the Silicon Valley Astronomy Lectures in the Smithwick Theater, Foothill College, El Monte Road and Freeway 280, in Los Altos Hills, California.

Free and open to the public. Parking on campus costs \$2.

Call the series hot-line at 650-949-7888 for more information and driving directions.

In 1998, observations of very distant exploding stars provided intriguing evidence that the expansion of the entire Universe is speeding up with time, rather than slowing down due to gravity as expected.

Today, new and completely independent observations strongly support this amazing conclusion. Over the largest scales of space, our Universe seems to be dominated by a repulsive "dark energy" stretching the very fabric of space itself.

Dr. Filippenko, who is a leader in the group that has made some of these remarkable observations, will give us a progress report on our "runaway universe." He is Professor of Astronomy at U.C. Berkeley, has written about 500 papers in astronomical publications, and has been voted the "Best Professor" on campus five times. He has been featured in three astronomy video courses published by The Teaching Company, and won the 2004 Carl Sagan Prize for Science Popularization.



The Pluto Exception

the list of planets in this solar system. Feel free to use the new definition to keep Ceres, Charon and Xena from becoming planets but leave Pluto as is.

Would this violate some deep scientific principle? If so, astronomy has plenty of precedent. Betelgeuse is listed as Alpha Orionis even though Rigel is the brightest star in Orion. M40, in the Messier list of "fuzzies" that could be comets, is actually just a double star but we don't remove it from the list (and there was no trouble in removing M102).

And there are still other things that make Pluto unique even today. Some of these are scientific and some are cultural. The scientific unique attributes are: discovered 70 years before another

object of the same size was discovered, 3 moons, and a barycenter that is not within Pluto. The cultural attributes include: named by an 11-year old girl, coincidentally named the same as the popular Disney character in the same year, and part of numerous mnemonic devices such as My Very Educated Mother Just Served Us Nine Pizzas.

So I think we have reason and precedent for declaring Pluto an exception, an exceptional planet. And I think if the powers-that-be in this debate didn't have to worry about Pluto's inclusion/exclusion, we would get a clear definition that would not require years of debate.

– PK

Deadly Planets

Patrick L. Barry and Dr. Tony Phillips

About 900 light years from here, there's a rocky planet not much bigger than Earth. It goes around its star once every hundred days, a trifle fast, but not too different from a standard Earth-year. At least two and possibly three other planets circle the same star, forming a complete solar system.

Interested? Don't be. Going there would be the last thing you ever do.

The star is a pulsar, PSR 1257+12, the seething-hot core of a supernova that exploded millions of years ago. Its planets are bathed not in gentle, life-giving sunshine but instead a blistering torrent of X-rays and high-energy particles.

"It would be like trying to live next to Chernobyl," says Charles Beichman, a scientist at JPL and director of the Michelson Science Center at Caltech.

Our own sun emits small amounts of pulsar-like X-rays and high energy particles, but the amount of such radiation coming from a pulsar is "orders of magnitude more," he says. Even for a planet orbiting as far out as the Earth, this radiation could blow away the planet's atmosphere, and even vaporize sand right off the planet's surface.

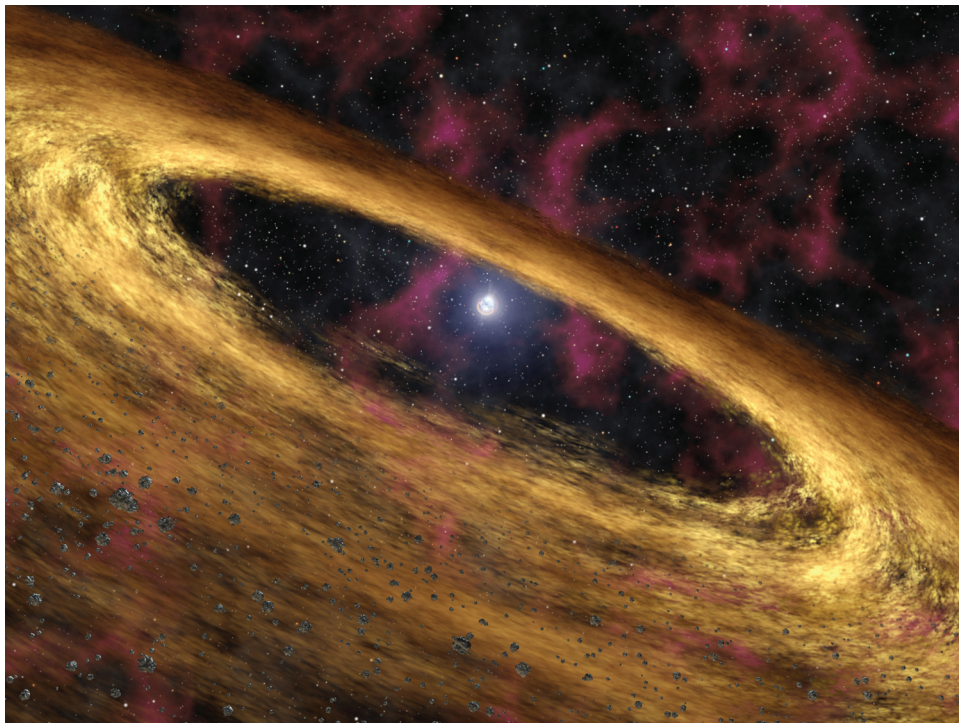
Astronomer Alex Wolszczan discovered planets around PSR 1257+12 in the

1990s using Puerto Rico's giant Arecibo radio telescope. At first, no one believed worlds could form around pulsars—it was too bizarre. Supernovas were supposed to destroy planets, not create them. Where did these worlds come from?

NASA's Spitzer Space Telescope may have found the solution. Last year, a group of astronomers led by Deepto Chakrabarty of MIT pointed the infrared telescope toward pulsar 4U 0142+61.

incoming radiation, creating colorful auroras across the sky. And though the pulsar would only appear as a tiny dot in the sky (the pulsar itself is only 20-40 km across), it would be enshrouded in a hazy glow of light emitted by radiation particles as they curve in the pulsar's strong magnetic field.

Wasted beauty? Maybe. Beichman points out the positive: "It's an awful place to try and form planets, but if you can do it there, you can do it anywhere."



Artist's concept of a pulsar and surrounding disk of rubble called a "fallback" disk, out of which new planets could form.

Data revealed a disk of gas and dust surrounding the central star, probably wreckage from the supernova. It was just the sort of disk that could coalesce to form planets!

As deadly as pulsar planets are, they might also be hauntingly beautiful. The vaporized matter rising from the planets' surfaces could be ionized by the

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

More news and images from Spitzer can be found at <http://www.spitzer.caltech.edu/>. In addition, The Space Place Web site features a cartoon talk show episode starring Michelle Thaller, a scientist on Spitzer. Go to <http://spaceplace.nasa.gov/en/kids/live/> for a great place to introduce kids to infrared and the joys of astronomy.

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Submit

Submit articles for publication in the SJAA Ephemeris. Send articles to the editors via e-mail to ephemeris@sjaa.net. **Deadline, 10th of previous month.**

SJAA loaner scope status

All scopes are available to any SJAA member; contact Mike Koop by email (koopm@best.com) or by phone at work (408) 473-6315 or home (408) 446-0310 (Please leave message, phone screened).

Available scopes

These are scopes that are available for immediate loan, stored at other SJAA members homes. If you are interested in borrowing one of these scopes, please contact Mike Koop for a scope pick up at any of the listed SJAA events.

# Scope	Description	Stored by
1	4.5" Newt/ P Mount	Annette Reyes
3	4" Quantum S/C	Hsin I. Huang
6	8" Celestron S/C	Karthik Ramamurthy
7	12.5" Dobson	Tom Fredrickson
10	Star Spectroscope	Jim Albers
11	Orion XT6 Dob	Ravi Shankar Erram
12	Orion XT8 Dob	Sarah E. Jones
13	Orion XT6 Dob	Rajiv Vora
14	8" f/8.5 Dob	Colm McGinley
15	8" f/9 Dobson	Mike Koop
19	6" Newt/P Mount	Daryn Baker
23	6" Newt/P Mount	Wei Cheng
24	60mm Refractor	Al Kestler
26	11" Dobson	Vivek Kumar
27	13" Dobson	Steve Houlihan
29	C8, Astrophotography	Mark Ziebarth
32	6" f/7 Dobson	Sandy Mohan
34	Dynamax 8" S/C	Yuan-Tung Chin
35	Meade 8" Equatorial	Mike Horzewski
38	Meade 4.5" Digital Newt	Tej Kohli
39	17" Dobson	Steve Nelson
40	Super C8+	Bill Kerns
42	11x80 Binoculars	Ritesh Vishwakarma
43	Orion XT4.5 Dob	Gary Mitchell

Scope loans

These are scopes that have been recently loaned out. If you are interested in borrowing one of these scopes, you will be placed on the waiting list until the scope becomes available after the due date.

# Scope	Description	Borrower	Due Date
8	14" Dobson	Charles Santori	10/2/06
28	13" Dobson	Craig Scull	8/1/06
33	10" Deep Space Explorer	Art Kalb	9/30/06
36	Celestron 8" f/6 Skyhopper	Kristi Whitfield	9/16/06
41	18" Sky Designs Dob	Len Bradley	9/17/06

Extended scope loans

These are scopes that have had their loan period extended. If you are interested in borrowing one of these scopes, we will contact the current borrower and try to work out a reasonable transfer time for both parties.

# Scope	Description	Borrower	Due Date
2	6" f/9 Dob	John Paul De Silva	?
9	C-11 Compustar	Bill Maney	Indefinite
16	Solar Scope	Mike Koop	Repair
21	10" Dobson	Michael Dajewski	Repair
37	4" Fluorite Refractor	Peter Young	8/11/06
44	4.5" Skyview/ P Mount	Mantle Yu	8/03/06

Waiting list: Empty! Reserve a scope today!

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☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

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